Suraj Kumar

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Education

Indian Institute of Information Technology Pune **Aug 2024 – May 2028**
*Bachelor of Technology in Electronics and Communication Engineering (**CGPA: 9.07**)* *Pune, Maharashtra*

Indian Institute of Information Technology Pune **Aug 2026 – May 2027**
Honours in Artificial Intelligence *Pune, Maharashtra*

Chinmaya Vidyalaya, Bokaro Steel City **2022 – 2024**
*Higher Secondary – Physics, Chemistry, Mathematics (**Grade: 91%**)* *Bokaro, Jharkhand*

Jawahar Navodaya Vidyalaya, Puto **2017 – 2022**
*Secondary Education (**Grade: 96.6%**)* *Koderma, Jharkhand*

Technical Skills

Languages: C++, Python, C, JavaScript, TypeScript, SQL, Assembly
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend & Frameworks: Node.js, Express.js, RESTful APIs, WebSockets, JWT, Streamlit
Databases: MongoDB, Redis, MySQL, Neo4j
Developer Tools: Git, GitHub, Docker, Postman, Linux/Ubuntu, Grafana, VS Code
AI/ML & Libraries: Deep Learning, TensorFlow, Keras, Pandas, NumPy, Computer Vision (CNN), LLM

Experience

Open Source Contributor | *drawDB (**39k+ stars** on GitHub)* | Debate AI (**Under AOSSIE-Organization**) **2026**
- Contributed to **drawDB** by engineering an **auto-connect foreign key feature** that dynamically infers relational schema links based on naming conventions.
- Contributed to **DebateAI**, an intelligent debating application powered by a high-performance Go backend and a modern TypeScript frontend.

Projects

Campus-Sphere | *React, Node.js, MongoDB, WebSocket, Redis, Docker, LLM, RAG* – Code | Live **June 2025**
- Built a scalable campus event ecosystem managing real-time state for concurrent active users using **React** and **Node.js**.
- Architected a real-time sports scoring engine using **Socket.IO** and async **MongoDB** writes, slashing broadcast latency to less than **100ms** across concurrent users.
- Engineered a contextual **RAG pipeline** with **Gemin API** and cosine similarity, caching 768-dim vector embeddings in **Redis** (7-day TTL) to reduce API calls by more than **80%**.
- Secured APIs with **Google OAuth** and rotating **JWTs**, monitored telemetry via **Prometheus**/**Grafana** and sustained **500 concurrent VUs** at less than **500ms p95** latency using **k6**.

OptimRoute | *Node.js, Next.js, Neo4j* – Code **July 2026**
- Modeled a transit network of **8,150+ stations** and rail segments as a directed graph using **Neo4j Graph Database**.
- Constructed optimized **Cypher queries** utilizing **CALL subqueries** to compute multi-hop transit paths evaluated by speed, distance, and transfer cost.

Achievements

- **Competitive Programming:** Solved **1000+ algorithmic problems**; **3-Star Coder** on CodeChef; Max Rating **1527 (Specialist)** on Codeforces , 1818 on LeetCode.
- **JEE Main 2024:** Secured **97.79 percentile** and **AIR 21624** in JEE Advanced.
- **Academic Selection:** Selected for **Jawahar Navodaya Vidyalaya**.
- **Scholarship:** Awarded the **National Scholarship Portal (NSP) Merit Scholarship**.

Coursework

Data Structures and Algorithms | Object-Oriented Programming | Database Management Systems (DBMS) | Operating Systems | Artificial Intelligence | Machine Learning | Digital Image Processing | Microprocessors